

LEARN MORE ABOUT DEEP LEARNING APPROACHES FOR PNEUMONIA DETECTION

A Comparative Study on Chest X-Ray Images

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CONTEXT AND MOTIVATION

Why should we use AI to detect pneumonia?



PNEUMONIA

- Top 10 cause of death globally;
- Accounting for over 15% of deaths in children under 5 worldwide;
- Difficult to diagnose early and consistently with traditional methods.

Deaths from pneumonia, by age, World, 1980 to 2021

The estimated annual number of deaths from pneumonia and other lower respiratory infections.

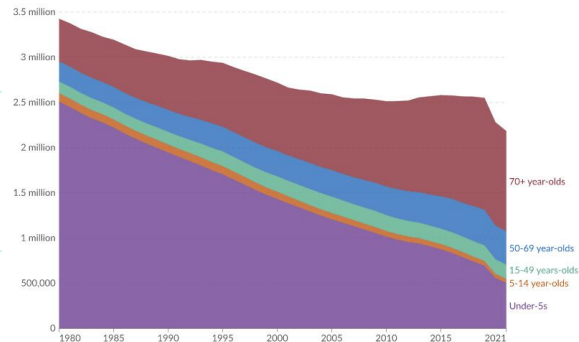
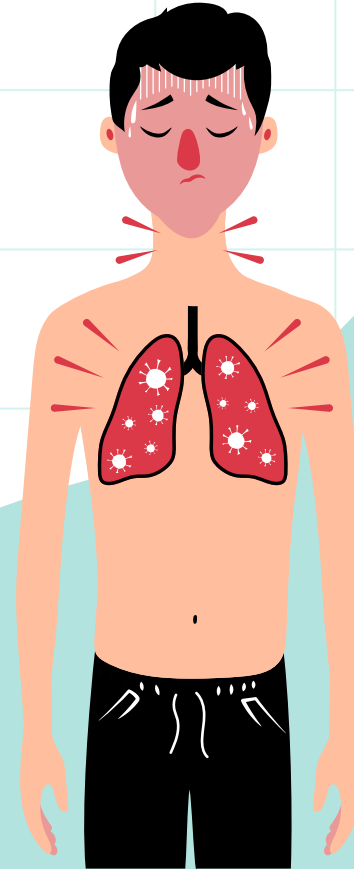


Fig. 1. Evolution of Deaths by Pneumonia



CHALLENGES IN DIAGNOSIS

- **Variable presentation across patients:** manifest differently depending on its severity and stage;
- **Overlapping signs with other diseases** (e.g., COPD, lung cancer);
- **Poor image quality and lack of historical comparisons;**
- **Low sensitivity in clinical assessments** (e.g., only 29%): doctors do not always agree on pneumonia predictions.

MOTIVATION

- AI and deep learning can support radiologists with more consistent, early, and scalable diagnoses.

PROBLEM STATEMENT



DESPITE THE POTENTIAL OF DL, SEVERAL CHALLENGES PERSIST

1. Many existing models fail when applied to unseen clinical environments or limited datasets.
2. Need to understand which pre-trained networks work best and under what conditions.



OBJECTIVES & CONTRIBUTIONS

1. Evaluate pre-trained CNNs for binary pneumonia detection.
2. Analyze the effect of data augmentation.
3. Explore transfer learning effectiveness in medical imaging.
4. Benchmark against existing datasets like MedMNIST.

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KEY DATASETS FOR PNEUMONIA DETECTION

MedMNIST:

- 18 standardized biomedical datasets (2D/3D)
- Fixed train/val/test splits → fair model comparison

Kaggle Pneumonia X-ray Dataset:

- Pediatric chest X-rays (ages 1–5)
- Expert-labeled: Normal, Bacterial, Viral Pneumonia

DEEP LEARNING MODELS IN MEDICAL IMAGE CLASSIFICATION

CheXNet (DenseNet-121)

- Outperformed radiologists (avg. F1 0.387 vs. CheXNet F1 0.43)
- Pioneered transfer learning in medical imaging

Vision Transformers (ViTs)

- Newer models surpass CNNs on accuracy/specificity

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PRE-TRAINED MODELS IN MEDICAL IMAGE CLASSIFICATION

Convolutional Neural Networks (CNNs)

Remain the core architecture for classifying chest X-ray images.

- ResNet (ResNet-18, ResNet-50):
 - Strong performance on MedMNIST (AUC 94–96%).
 - Commonly used with transfer learning (ImageNet).
- DenseNet (121, 169) - used in CheXNet:
 - Outperformed radiologists and is an excellent feature extraction for pneumonia tasks.
- VGG & MobileNet:
 - Simpler architectures, used in Kaggle implementations.
 - MobileNet is efficient for low-resource environments.
- EfficientNet:
 - High accuracy with fewer parameters.
 - Layer-freezing strategy improves transfer learning efficiency.

Advanced Models and Hybrid Approaches

- Vision Transformers (ViTs):
 - Capture long-range dependencies better than CNNs.
 - Achieved 97.6% accuracy and 95% sensitivity (Zhang et al., 2024).
 - Outperformed ResNet-50 on pediatric X-rays.
- Ensemble Models:
 - Combine multiple architectures (e.g., GoogLeNet + ResNet + DenseNet).
 - Improve generalization and stability.
 - PLOS ONE: Ensemble AUC 0.962 vs. 0.948 for best single model.

STATE OF ART

LIMITATIONS & RESEARCH GAPS

Evaluation Issues

Accuracy overemphasized - limited use of AUC, F1, recall.

Generalization Gaps

Most models trained and tested on only one dataset: fail to evaluate models across diverse datasets.

Class Imbalance

Few studies apply focal loss or data augmentation effectively.

Clinical Integration

Lack of interpretability.

Rare real-world validation with doctors or patient subgroups.



DATASET OVERVIEW

PNEUMONIA-MNIST DATASET

Subset of MedMNIST with 5,856 pediatric chest X-ray images

Pre-split into:

- Training: 4,708 images
- Validation: 524 images
- Test: 624 images

Two classes:

- Normal (27%)
- Pneumonia (73%)



Slight class imbalance towards Pneumonia

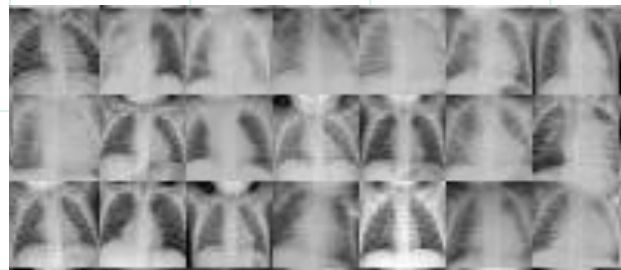


Fig. 2. Overview of Chest X-Ray Images from the chosen dataset

TABLE I
DATASET COMPOSITION AND CLASS DISTRIBUTION

Set	Total Images	Normal	Pneumonia	Percentage
Training	4708	1214	3494	80.4%
Validation	524	135	389	8.9%
Testing	624	234	390	10.7%
Total	5856	1583	4273	100%



IMAGE CHARACTERISTICS AND PREPROCESSING

PREPROCESSING PIPELINE

There are available in various resolutions: 28x28, 64x64, 128x128, and 224x224.

We follow the authors' recommendation that **28×28 inputs** yield similar or slightly lower AUC at a lower computation cost.

Steps applied:

1. Reshaping: images transformed to a consistent 28x28 pixel dimension with a single channel.
2. Pixel normalization to [0, 1] range.
3. Labels one-hot encoded for binary classification.





DATASET PREPROCESSING

DATA AUGMENTATION

Applied only to the **training set**.

Techniques used:

1. Random rotation ($\sim 15^\circ$).
2. Horizontal and vertical translation.
3. Random zoom.
4. Contrast adjustment.

It helps **prevent overfitting** and improves generalization.

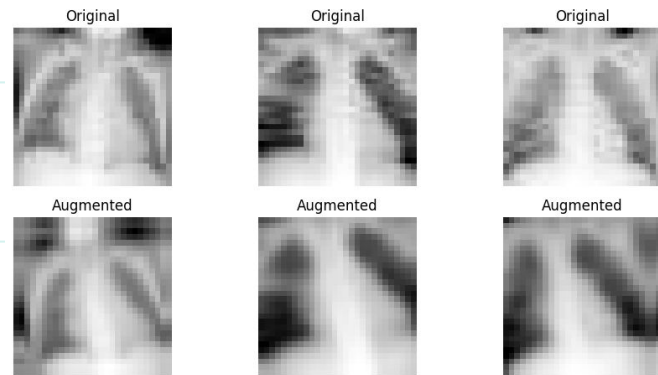


Fig. 3. Original vs Augmented Images

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METHODOLOGY OVERVIEW



OVERALL PIPELINE

Two-stage pipeline: **Preprocessing** and **Classification**.

- Implemented multiple CNN architectures.
- Applied transfer learning and regularization.
- Evaluated using standard classification metrics.

CLASSIFICATION

- AlexNet
- ResNet-18
- ResNet-50
- DenseNet-169 (with ImageNet transfer learning)

With and without data augmentation.

MODEL ARCHITECTURE

ALEXNET ARCHITECTURE

- 5 convolutional blocks + batch norm & ReLU
- 3 max-pooling layers
- 2 dense layers (4096 units each) + dropout
- Final softmax output for binary classification

Adjusted strides/padding for 28×28 input

RESNET-18 AND RESNET-50

ResNet-18:

- Modified initial kernel (3×3, stride 1)
- 4 stages × 2 residual blocks
- Global average pooling layer, Dropout for regularization, Dense layer with softmax activation

ResNet-50:

- Bottleneck blocks (1×1, 3×3, 1×1)
- Four stages with varying numbers of bottleneck blocks (3, 4, 6, and 3)
- Global average pooling, dropout, and final classification layers

DENSENET-169 WITH TRANSFER LEARNING

- Dense connections for feature reuse
- Grayscale to 3-channel conversion
- Resized to 224×224
- Pretrained on ImageNet with weights
- Custom classification head: GAP + dropout + softmax

TRAINING STRATEGY

Transfer Learning:

- Only DenseNet-169 used pre-trained weights
- Others trained from scratch
- All models trained end-to-end (no layer freezing)

Regularization:

- Batch Normalization
- Dropout (fixed and scheduled)
- Early Stopping (patience: 10)

OPTIMIZATION SETUP

TABLE II
HYPERPARAMETER SETTINGS AND RANGES EXPLORED

Hyperparameter	Values Explored
Optimizer	Adam
Loss Function	Binary Cross-Entropy
Initial Learning Rate	1×10^{-3}
LR Reduction Factor	0.2 (on validation plateau)
Batch Size	32, 64, 96, 100
Dropout Strategy	Fixed (0.3, 0.5); Scheduled (0.5→0.3)
Data Augmentation	On / Off (training set only)
Maximum Epochs	100
Early Stopping Patience	10
Model Selection Criterion	Validation AUC

EVALUATION METRICS

ACCURACY

Overall correctness

AUC

Class separation ability

PRECISION & RECALL

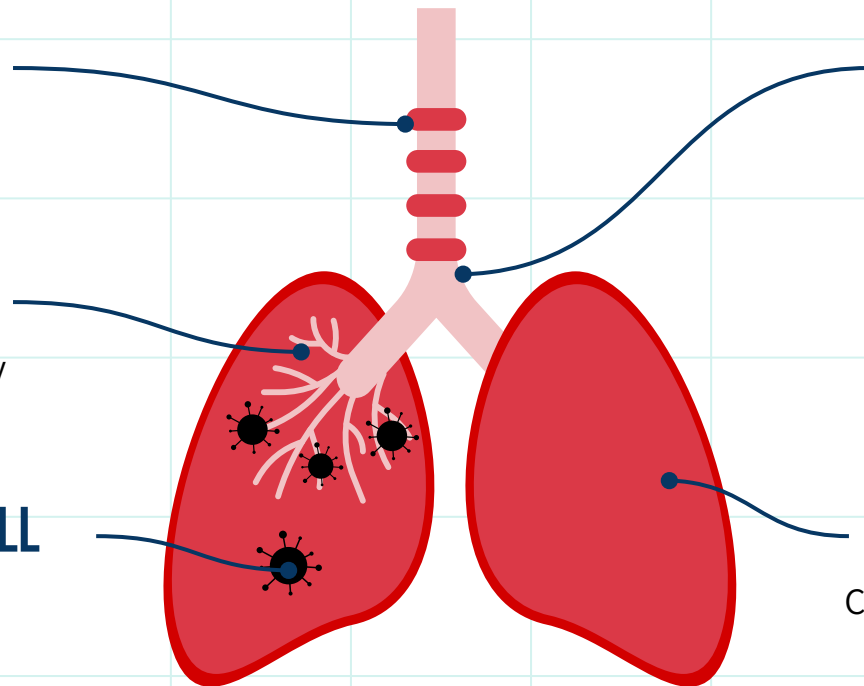
Medical diagnostic importance

F1-SCORE

Balance between precision & recall

CONFUSION MATRIX

Classification breakdown





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RESULTS AND DISCUSSION

Summary of Best Results by Model and Data
Augmentation Impact

OVERALL MODEL PERFORMANCE

SUMMARY OF BEST RESULTS BY MODEL

Best overall: ResNet-50 with augmentation

- AUC: 0.9791, Accuracy: 88.6%, FNR: 0%

AlexNet also achieved **0% FNR**, AUC: 0.9704, lower accuracy (85.6%)

ResNet-18 and DenseNet-169: FNR: 0.26% (**1 false negative each**)

TABLE III
BEST CLASSIFICATION RESULTS FOR EACH MODEL ARCHITECTURE

Model	FNR	FN Count	AUC	Acc.
AlexNet	0.00%	0	0.9704	0.8558
ResNet-18	0.26%	1	0.9736	0.8798
ResNet-50	0.00%	0	0.9791	0.8862
DenseNet-169	0.26%	1	0.9514	0.8077

AUC COMPARISON

Augmentation improved AUC across all models

- ResNet-50: **AUC increased** from $\sim 0.91 \rightarrow \sim 0.97$

Most top-performing models used augmentation

Dropout strategy had **minimal AUC impact** compared to augmentation

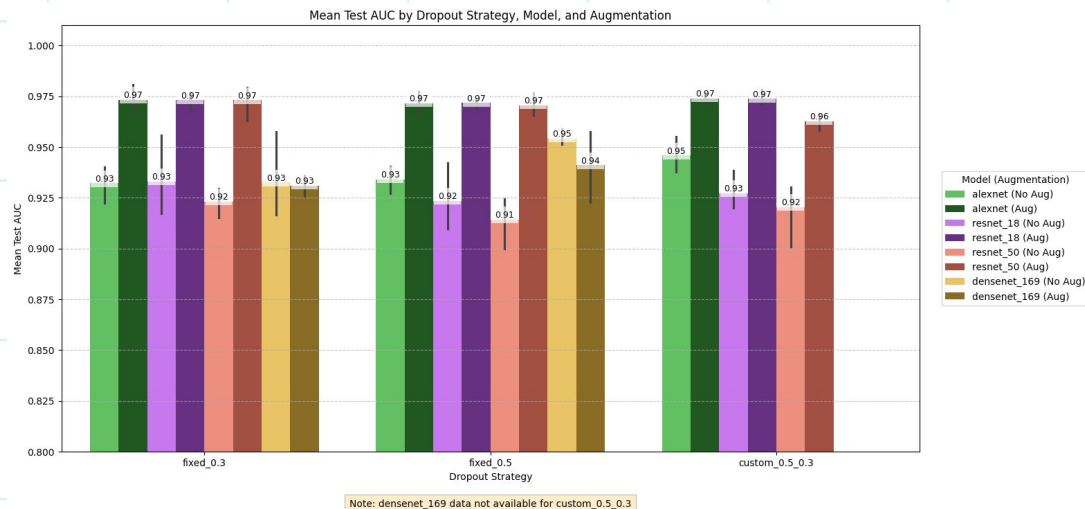
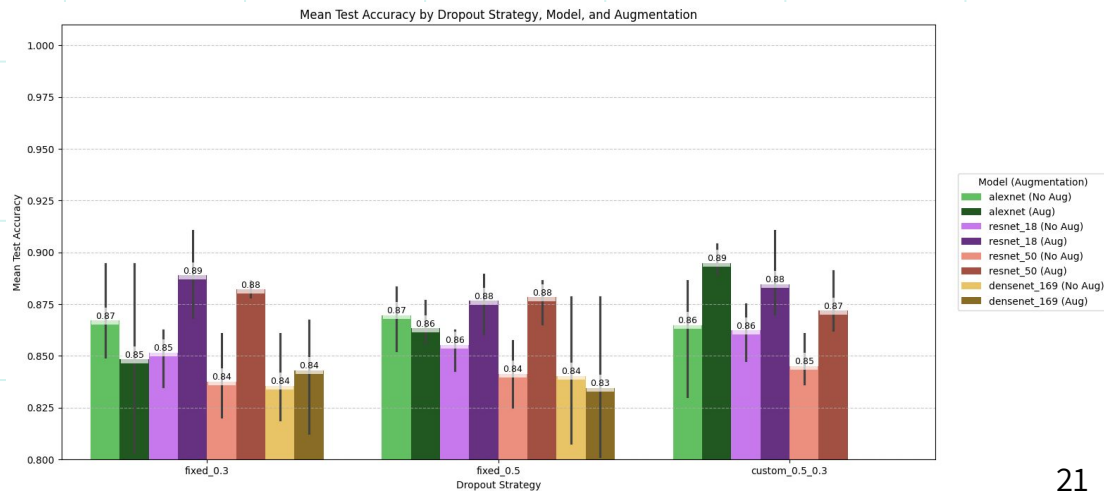


Fig. 4. Mean Test AUC by Dropout Strategy, Model, and Augmentation

ACCURACY COMPARISON

- **ResNet-18 + augmentation** - highest accuracy: ~91%
- **AlexNet + augmentation** - up to 89.4% with correct tuning
- **DenseNet-169** consistently **underperformed** ($\leq 84\%$) despite complexity

Fig. 5. Mean Test Accuracy by Dropout Strategy, Model, and Augmentation



COMPUTATIONAL EFFICIENCY

EXECUTION TIME PER MODEL



Fig. 6. Mean Execution Time by Batch Size, Model, and Augmentation

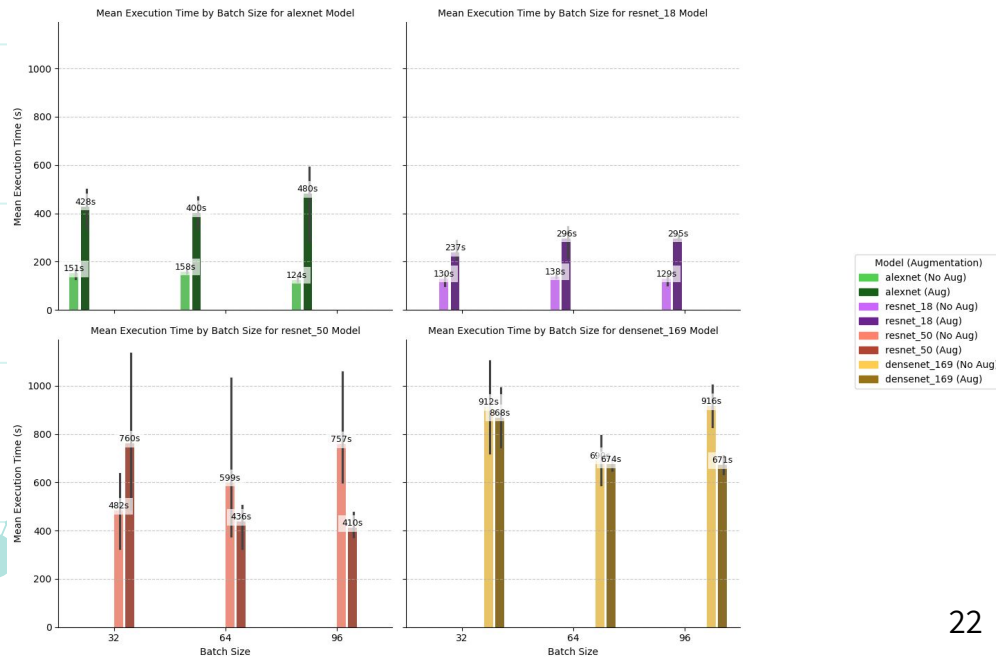
Mean Execution Time by Batch Size, Model, and Augmentation

DenseNet-169 - slowest model (671–916s per epoch)

ResNet-50: moderate speed, but variable depending on batch size

AlexNet & ResNet-18: most efficient models

Augmentation increases training time up to +200%



BEST CONFIGURATIONS BY FNR

TOP 5 MODELS – RANKED BY FALSE NEGATIVE RATE

Top models with 0% FNR:

- ResNet-50 (aug, batch 32, dropout 0.3)
- AlexNet (aug, batch 96, dropout 0.5)

Data augmentation and smaller batch sizes dominate top ranks.

Curiously, DenseNet only shows up in 9th place even when considering top 20 models.

TABLE IV
TOP 5 MODELS BY FALSE NEGATIVE RATE

Model	Aug.	Batch	Dropout	FNR	AUC	Acc.
ResNet-50	Yes	32	Fixed (0.3)	0.00%	0.9791	0.8862
AlexNet	Yes	96	Fixed (0.5)	0.00%	0.9704	0.8558
ResNet-50	No	64	Fixed (0.5)	0.00%	0.8998	0.8253
AlexNet	Yes	32	Fixed (0.3)	0.26%	0.9805	0.8478
AlexNet	Yes	64	Fixed (0.3)	0.26%	0.9773	0.8942

BEST CONFIGURATIONS BY AUC

TOP 5 MODELS – RANKED BY AUC

AlexNet (aug, batch 32, dropout 0.3) - AUC: 0.9805

ResNet-50 and ResNet-18 also among the top

Caution: High AUC $\neq$ low FNR can lead to **clinical risk**
if FNR is ignored

DenseNet is not present in top 25 results.

TABLE V
TOP 5 MODELS BY AUC SCORE

Model	Aug.	Batch	Dropout	AUC	Acc.
AlexNet	Yes	32	Fixed (0.3)	0.9805	0.8478
ResNet-50	Yes	32	Fixed (0.3)	0.9791	0.8862
ResNet-18	Yes	32	Custom (0.5-0.3)	0.9781	0.9103
AlexNet	Yes	32	Fixed (0.5)	0.9773	0.8766
AlexNet	Yes	64	Fixed (0.3)	0.9773	0.8942

KEY OBSERVATIONS

KEY INSIGHTS FROM MODEL COMPARISON

ResNet-50 offers **best clinical safety + performance trade-off**

AlexNet shows **strong performance when optimized** - lightweight + accurate

DenseNet-169 **underperformed** across all metrics

Clinical context favors low FNR, not just high AUC

IMPACT OF DATA AUGMENTATION

- +0.034 AUC improvement on average
- **Accuracy increase** from 0.852 → 0.870
- But **FNR increase** from 0.68% → 1.24%
- **Execution time increase** by ~20%

Trade-off: **better generalization vs. increased false negatives**

TABLE VI
AVERAGE EFFECT OF DATA AUGMENTATION

Augmentation	Acc.	AUC	FNR	Exec. Time
With Augmentation	0.870	0.965	1.24%	474.8 s
Without Augmentation	0.852	0.931	0.68%	395.7 s

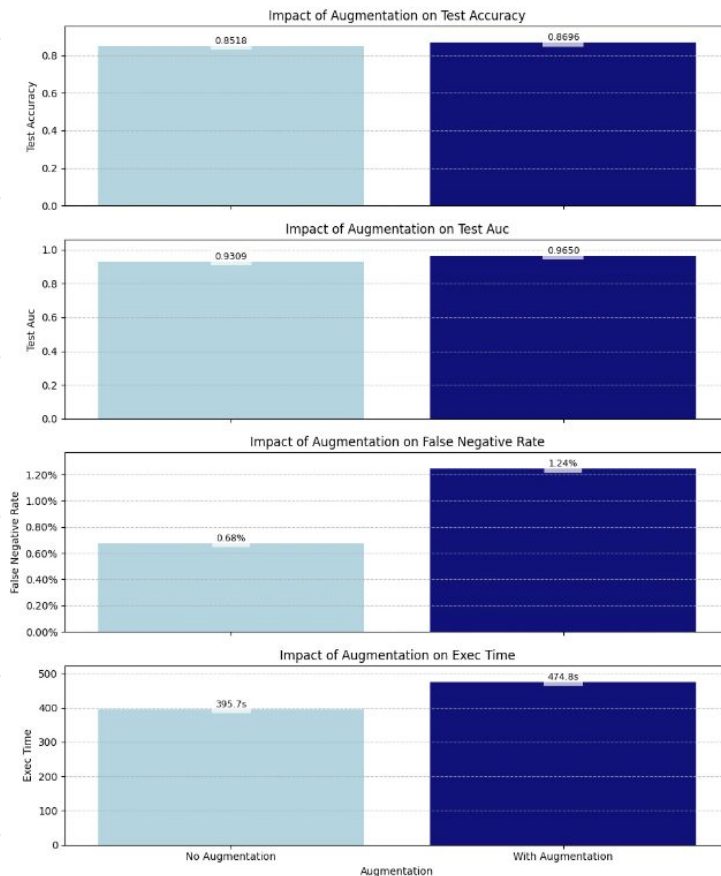


Fig. 7. Impact of Data Augmentation on Model Performance

BATCH SIZE ANALYSIS

BATCH SIZE VS. PERFORMANCE

Smaller batch (32): best accuracy (0.873) and AUC (0.953)

Larger batch (96): lowest FNR (0.72%)

Batch 64 offers best balance: solid metrics + faster training

Smaller batches - better optimization, more updates

TABLE VII
IMPACT OF BATCH SIZE ON MODEL PERFORMANCE

Batch Size	Acc.	AUC	FNR	Exec. Time
32	0.873	0.953	1.27%	460.6 s
64	0.860	0.946	0.89%	400.9 s
96	0.849	0.945	0.72%	444.2 s

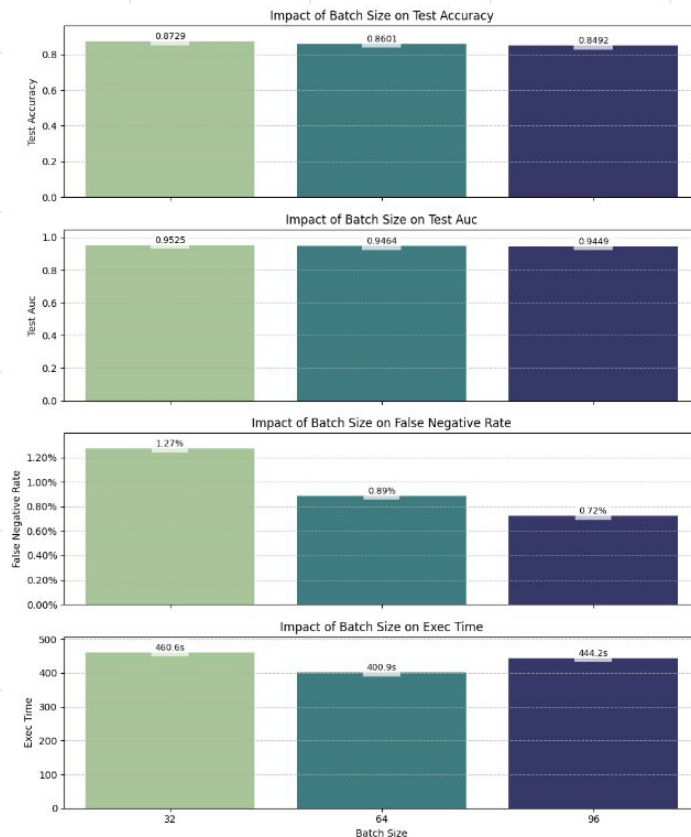


Fig. 8. Impact of Batch Size on Model Performance

DROPOUT STRATEGY COMPARISON

Best overall: Custom dropout (0.5 $\rightarrow$ 0.3) with highest AUC (0.951), best accuracy (0.871)

Lowest FNR: Fixed dropout (0.3) with FNR = 0.84%

Custom strategy also fastest (9% less time than fixed 0.3)

Adaptive dropout helps balance learning and overfitting

TABLE VIII
EFFECT OF DIFFERENT DROPOUT STRATEGIES

Dropout Strategy	Acc.	AUC	FNR	Exec. Time
Custom (0.5, 0.3)	0.871	0.951	1.07%	420.8 s
Fixed (0.3)	0.857	0.946	0.84%	463.6 s
Fixed (0.5)	0.857	0.948	0.99%	417.7 s

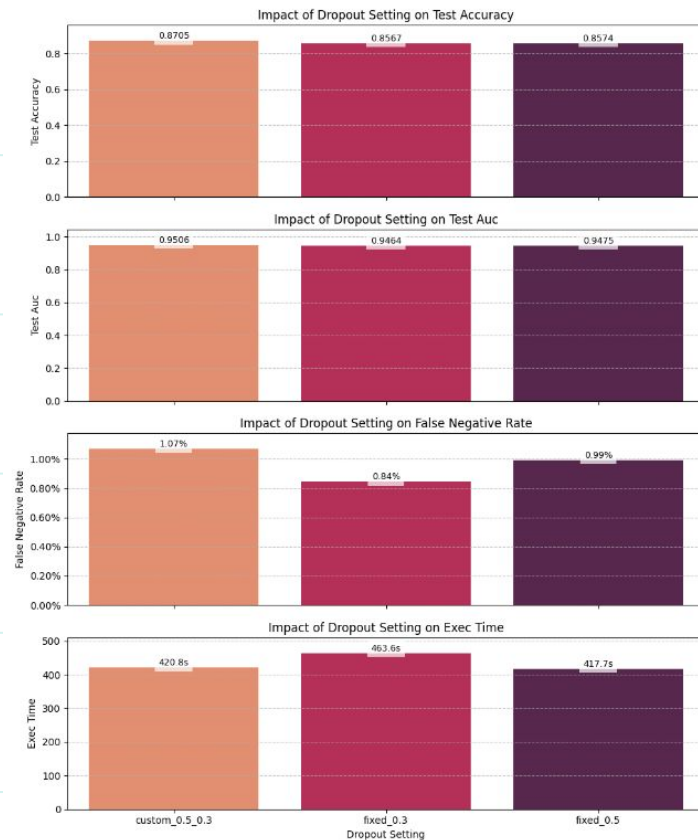


Fig. 9. Impact of Dropout Strategy on Model Performance

QUALITATIVE INSIGHTS

ARCHITECTURE BEHAVIOR OBSERVATIONS

AlexNet strong despite simplicity.

ResNet-18 & AlexNet benefited most from **augmentation**.

Shallow models + good regularization $\approx$ deep models.

We conclude that **architecture is not everything** and **optimization matters**.

FALSE NEGATIVE ANALYSIS

DenseNet-169 (no augmentation): lowest FNR = 0.47%

Augmentation increase FNR in all models (e.g., DenseNet-169 has 2.26%)

AlexNet had minor FNR increase; others were more affected.

Key takeaway: Augmentation boosts AUC but can hurt sensitivity.

TABLE IX
FALSE NEGATIVE RATES BY MODEL AND AUGMENTATION STRATEGY

Model	Without Augmentation	With Augmentation
AlexNet	0.71%	0.60%
ResNet-18	0.68%	1.14%
ResNet-50	0.77%	1.31%
DenseNet-169	0.47%	2.26%

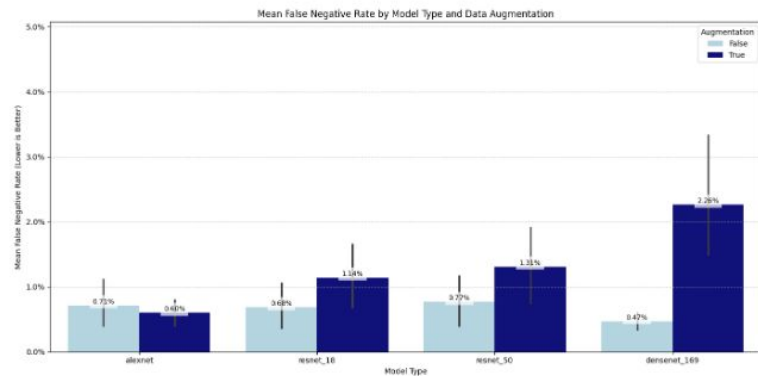


Fig. 10. False Negative Rate Comparison Across Models

CORRELATION INSIGHTS

- Accuracy $\leftrightarrow$ Precision (0.98), F1 (0.99)
- AUC $\leftrightarrow$ Precision (0.55), but $\leftrightarrow$ Recall (0.01) $\rightarrow$ AUC $\neq$ Sensitivity
- FNR $\leftrightarrow$ Recall (-0.96), $\leftrightarrow$ Precision (+0.37)
- Execution time weakly correlated with performance
- Smaller batch sizes $\rightarrow$ better accuracy, precision, F1
- Larger batch sizes $\rightarrow$ better recall ($\uparrow$ sensitivity)

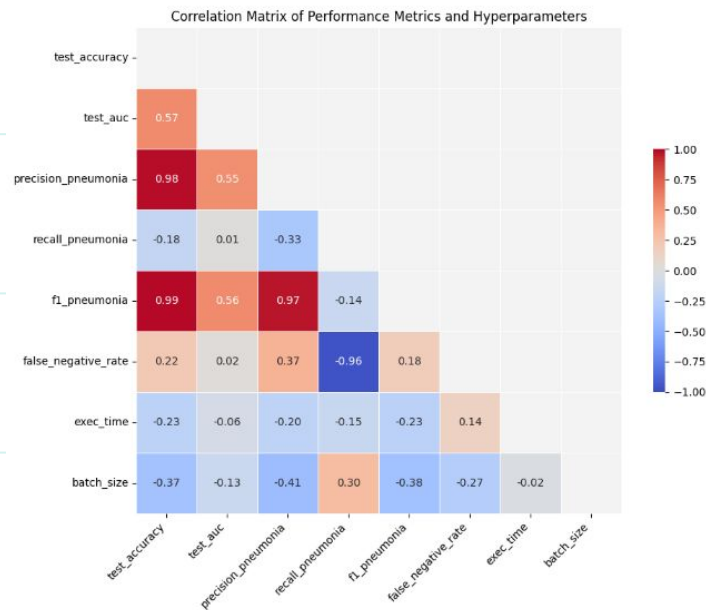


Fig. 11. False Negative Rate Comparison Across Models

DATA AUGMENTATION TRADE-OFFS

AUGMENTATION: PERFORMANCE VS. COST



- +3.4% AUC gain on average with augmentation
- Training time increase by ~20%

- **Best gains:**

- ResNet-50: +5.0% AUC
- ResNet-18: +4.5% AUC

- **DenseNet-169:** AUC ↓ 0.7% with augmentation
 - Augmentation should be architecture-dependent

TABLE X
AUC PERFORMANCE BY MODEL AND AUGMENTATION STRATEGY

Model	Without Aug.	With Aug.	Improvement
AlexNet	0.937	0.973	+3.6%
ResNet-18	0.928	0.973	+4.5%
ResNet-50	0.919	0.969	+5.0%
DenseNet-169	0.943	0.936	-0.7%

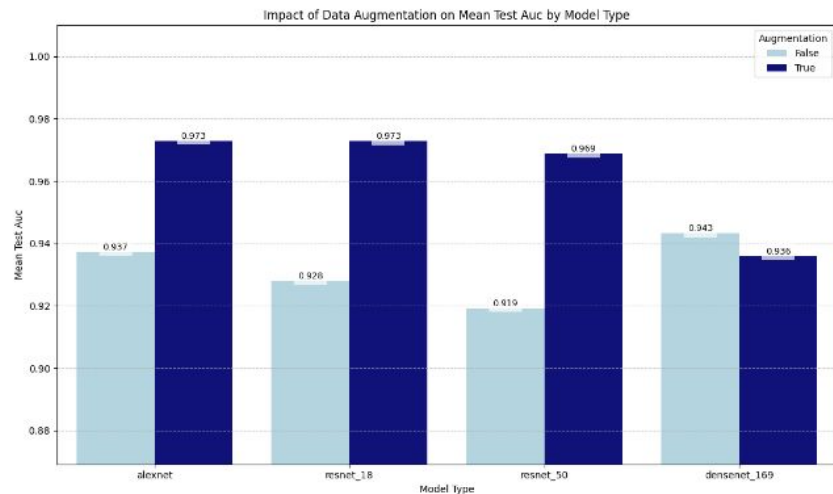


Fig. 12. Impact of Data Augmentation by Model Architecture

FALSE NEGATIVES VS. FALSE POSITIVES

CLINICAL TRADE-OFF: FNR VS. FPR

- Lowering FNR usually increases FPR
- **ResNet-50 (Aug):**
 - 0% FNR at ~30-35% FPR → best trade-off
- **AlexNet (Aug):**
 - Also achieved near-0% FNR with consistent results
- **DenseNet-169 (Aug):** worst trade-off (FNR > 4.5%)
- Clinical priority = **minimize FNR**, even if FPR is higher

TABLE XI
MODEL PERFORMANCE AT LOW FNR ($\leq 0.5\%$) FOR PNEUMONIA
DETECTION

Model	Approximate FPR at FNR $\leq 0.5\%$
ResNet-50 (Aug)	30-35%
AlexNet (Aug)	35-40%
ResNet-18 (Aug)	35-45%
DenseNet-169 (Aug)	40-50%

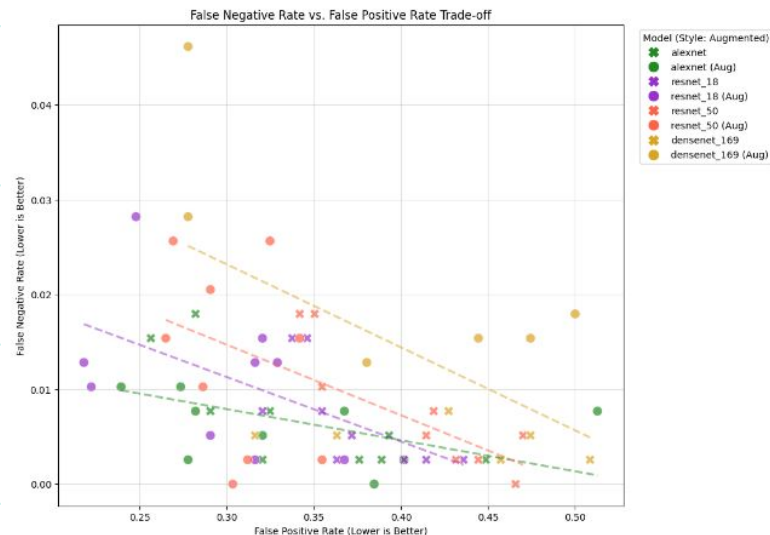


Fig. 13. False Negative Rate vs. False Positive Rate Trade-off for Pneumonia Detection

EFFICIENCY TRADE-OFFS

AUC VS. TRAINING TIME

- **ResNet-18 (Aug):** best balance of AUC and speed
- **ResNet-50:** high AUC, but 3–4× longer training
- **DenseNet-169:** high cost, low gain
- Simpler models (e.g. AlexNet) are **more efficient**

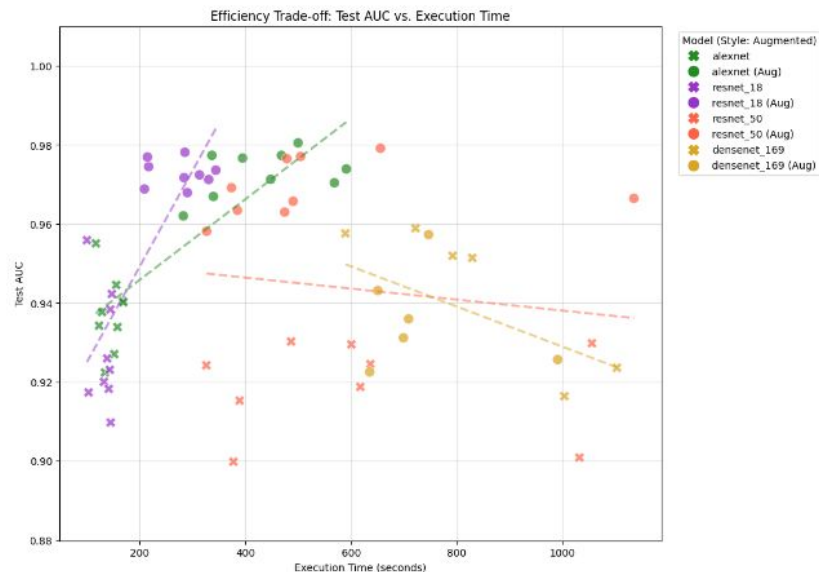


Fig. 14. Efficiency Tradeoff: AUC vs. Training Time

BENCHMARK COMPARISON

COMPARISON WITH EXISTING BENCHMARKS

Why We Outperformed Benchmarks?

1. Extensive Data Augmentation → improved generalization
2. Careful Hyperparameter Tuning → better convergence
3. Focus on FNR Reduction → clinically relevant optimization
4. Smarter Model Selection: balancing complexity and utility



TABLE XII
COMPARISON OF OUR RESULTS WITH LITERATURE BENCHMARKS

Model	Accuracy (%)	AUC	Source
ResNet-18 (28)	85.4	0.944	MedMNIST
ResNet-18 (224)	86.4	0.956	MedMNIST
ResNet-50 (28)	85.4	0.948	MedMNIST
ResNet-50 (224)	88.4	0.962	MedMNIST
auto-sklearn	85.5	0.942	MedMNIST
Google AutoML Vision	94.6	0.991	MedMNIST
ResNet-18 (Ours)	87.98	0.974	This work
ResNet-50 (Ours)	88.62	0.979	This work
AlexNet (Ours)	85.58	0.970	This work
DenseNet-169 (Ours)	80.77	0.951	This work
CheXNet (DenseNet-121)	76.8	0.897	Rajpurkar et al.

CONCLUSION

SUMMARY OF FINDINGS

Best model: ResNet-50 with augmentation – AUC 0.9791, 0% FNR

AlexNet: Simple yet effective – AUC 0.9704, 0% FNR

Data augmentation: Boosted AUC by +0.034 avg

- Notable gains: +4.5% (ResNet-18), +5.0% (ResNet-50)

Hyperparameters:

- Best batch size: 32.
- Custom dropout: 0.5 early / 0.3 late.
-

Outperformed MedMNIST benchmarks:

- Strong **focus** on **false negative minimization**.

CONCLUSION

LIMITATIONS

High false positives (25–50%): May cause unnecessary follow-ups.

Augmentation trade-off: DenseNet-169 FNR increased from 0.47% → 2.26%

Computational cost: DenseNet-169 expensive with limited gains.

Still behind Google AutoML Vision: Accuracy 94.6%, AUC 0.991 (Yang et al., 2021)

Binary classification only: Can't distinguish bacterial vs. viral pneumonia.

FUTURE WORK

TECHNICAL IMPROVEMENTS

Model Enhancements

- **Vision Transformers:** Better attention could lower FPR & FNR.
- **Ensemble methods:** Balance strengths (e.g., ResNet-50 + low-FPR models)
- **Custom loss functions:** Focal Loss for false negative penalty.
- **Threshold calibration:** Adjust decision threshold based on clinical context.

BROADER IMPACT

Practical Advancements

- **Two-phase transfer learning:** Freeze → fine-tune for better feature use.
- **Interpretability tools:** Grad-CAM to visualize decision areas.
- **Diverse dataset validation:** Improve generalization & detect bias.
- **Clinical integration:** Show FNR/FPR trade-offs to support doctor decisions.



THANKS!

Do you have any questions?